

Sovereign Debt Pricing with Long-Run Capacity and Short-Run Risk Appetite: A Minimal Equilibrium Model

Model note, v7 (essentials) — for review

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1 What this note does

This note states a minimal equilibrium model of sovereign debt pricing for a euro-area member state: a small open economy that cannot print the unit of account (in the sense of Bolton–Huang: no unilateral monetization). The model has three primitives, introduced in Section 2: the process of nominal growth; the state’s extraction capacity, with two components — the fiscal surplus, extracted from taxpayers, and the flow extracted from *captive creditors* — each bounded by a participation constraint; and the union safe rate R^f . Institutional detail enters the framework at exactly two points: as an extraction flow (a determinant of long-run capacity), or at the period- t auction (a determinant of what the market must absorb and at what price). All state-contingency runs through the auction: through the realized pair (G_t, θ_t) and the clearing default probability p_t . Extensions that would require state-contingent institutions are deferred; Section 9 records them.

The sections divide the work as follows. Section 3 analyzes short-run borrowing at date t for a *generic* function $P(\cdot)$ relating the default probability to the face value issued; everything that matters for multiple equilibria follows from two qualitative properties of P . Calendar time appears here, and only here. Section 4 develops P from first principles, following Collard–Habib–Rochet (2015; CHR), with the captive layer inside the capacity recursion; it ends with a neutrality proposition for anticipated union-wide inflation and a corollary isolating the one inflation variable that matters: the differential between domestic and union-wide inflation. Section 5 adds systematic risk: default happens in global bad states (Longstaff–Pan–Pedersen–Singleton, 2011; LPPS), so the probabilities in the pricing must be risk-neutral, governed by a short-run market price of distress risk θ_t and its long-run anchor θ_∞ . Section 6 translates fragility into risk-appetite space; Sections 7–8 give the limits recovering Blanchard’s (2022) box and CHR, and a calibrated illustration. Notation is independent of the sources (Appendix A); the relation to operational DSA practice is in Appendix B.

2 Setup

Time is annual. Nominal GDP grows at the gross rate $G_{t+1} = y_{t+1}/y_t$, i.i.d. lognormal:

$$\ln G_{t+1} \sim N(\mu, \sigma^2), \quad F(\cdot) \text{ its cdf}, \quad \bar{g} = \mathbb{E}[G] = e^{\mu + \sigma^2/2}. \quad (1)$$

Inflation. Nominal growth bundles real growth with domestic inflation, and the safe rate bundles the real rate with union-wide inflation. Write

$$\mu = \mu_r + \pi^d, \quad \ln R^f = r + \pi^u, \quad e \equiv \pi^d - \pi^u, \quad (2)$$

where π^d is expected domestic inflation, π^u expected union-wide inflation (to which the union safe asset is Fisher-linked), and e the expected inflation differential. The gross safe return R^f is set by the union-wide safe asset and exogenous to the small economy; π^d is not — a member state's inflation can move independently of π^u . Section 4.4 shows that the model is exactly neutral with respect to the common component and depends on the two inflation rates only through e .

Extraction capacity. The state can extract from its stakeholders a net resource flow bounded by participation constraints. Two components are modeled. From *taxpayers*: a primary surplus of at most $\bar{s}$ per year (share of GDP), bounded by the response of the tax base and the political limit on the surplus; the state follows a fiscal plan $\{s_t\}$, $s_t \leq \bar{s}$. From *captive creditors*:

Definition 1 (captive creditors). A captive creditor supplies proceeds c (per unit of current GDP) to every auction, at the accepted gross return $R^c = R^f/(1+w)$, $w \geq 0$, irrespective of the clearing default probability: the face value it rolls over is

$$k = \frac{c R^f}{1+w},$$

for which it pays the price $(1+w)/R^f$ per unit, where the marginal lender pays $q(p_t)$ of (4). One should think of the pair (w, c) as linked through a regulatory perimeter,

$$c \leq \bar{c}(w), \quad \bar{c} \text{ nonincreasing},$$

which formalizes leakage: captivity is produced by regulation (preferential treatment of sovereign exposures, portfolio mandates, moral suasion), written in value-ratio terms, and widening the wedge w shrinks the base it can be collected from. The face value k is kept as shorthand throughout.

For euro-area countries the empirically relevant case is $w \approx 0$ with $k > 0$: captivity today means risk-*insensitivity* (regulatory home bias of domestic banks and mandated funds), not below-safe returns. Sections 3 and 4 define the flows extracted from captive creditors through their overpayment for government debt issues relative to the marginal lender.

Debt. Sovereign debt is a one-period zero-coupon bond: d_t is the total face value due at $t+1$ per unit of date- t GDP (market and captive tranches together, $d_t \geq k$), R_t the gross promised return, $\psi_t = R_t - R^f$ the spread. In default, face values are written down by the haircut $h \in (0, 1)$; recovery is $1 - h$.

Lenders. The *marginal* lender is competitive and deep-pocketed; the pricing condition (4) refers to this lender. Captive creditors do not affect pricing at the margin; they affect the budget, through the tranche they absorb and the price they pay for it.

Timing of the period- t auction.

1. Growth G_t (and, from Section 5 on, the market's risk appetite θ_t) realizes. The inherited obligation per unit of current GDP is d_{t-1}/G_t .
2. The government runs the surplus s_t and must finance its **refinancing need**

$$n_t = \frac{d_{t-1}}{G_t} - s_t. \quad (3)$$

This is an accounting identity. With one-period debt the whole stock rolls each year; with a maturity structure, n_t is maturing debt plus interest minus the primary surplus — the *gross financing need* (GFN), a published statistic. The model's short-run state variable is observable.

3. The government offers total face value d_t . The captive creditors take face value k at their price (proceeds c); the market prices the residual tranche $d_t - k$ at q_t per unit of face value. Feasibility requires that total proceeds equal n_t . If no d_t can raise n_t , the government defaults today (write-down h).

Pricing a defaultable promise. A lender indifferent between the safe asset and a bond that pays in full with probability $1 - p$ and pays recovery $1 - h$ with probability p prices a unit of face value at its discounted expected payoff:

$$q(p) = \frac{1 - hp}{R^f}, \quad R = \frac{R^f}{1 - hp}, \quad \psi = \frac{R^f hp}{1 - hp} \approx hp. \quad (4)$$

Until Section 5, read p as a physical probability priced risk-neutrally; Section 5 shows that with systematic default risk p must be the *risk-neutral* probability, and how it is constructed.

Who knows what. The structure and parameters — including (w, c) — are common knowledge, and so is the function P introduced next. What no individual investor knows, and the government cannot choose, is the price at which today's auction will clear: equivalently the clearing spread, the promised return, or the default probability p_t (one conjecture, not four). Each investor prices against a conjecture about the marginal investor. This is the entire information structure; the multiple-equilibrium problem lives in it.

3 Short-run borrowing with a generic default-probability function

Suppose — deferring the microfoundation to Section 4 — that the probability of default at $t+1$ depends on the total face value issued at t through a function P :

Assumption 1 (default-probability function). $P : [0, \infty) \rightarrow [0, 1]$ is continuously differentiable and nondecreasing, with $P(0) = 0$ and $P(d) \rightarrow 1$ as $d \rightarrow \infty$.

Definition 2 (temporary equilibrium). Given the refinancing need $n_t \geq c$,¹ a temporary equilibrium at date t is a pair (d_t, p_t) such that (1) pricing: $q_t = (1 - hp_t)/R^f$; (2) budget: $q_t(d_t - k) + c = n_t$; (3) consistency: $p_t = P(d_t)$.

¹For $n_t < c$ the captive rollover alone covers the need and no market issuance is required; this corner is ignored throughout.

Substituting (1)–(2) into (3), an equilibrium is a solution d of one scalar equation,

$$\boxed{\xi(d) \equiv \frac{(1 - hP(d))(d - k)}{R^f} + c = n_t} \quad (5)$$

where ξ is the **funding curve**: total proceeds as a function of total face value offered. It decomposes into the marginal lender's valuation of the whole issue and the captive holders' overpayment,

$$\xi(d) = q(P(d))d + \alpha_t(d), \quad \alpha_t(d) \equiv c - q(P(d))k = \frac{k(w + hP(d))}{R^f}, \quad (6)$$

the **realized extraction flow**: the wedge w plus the risk transfer $hP(d)$, per unit of captive stock. Note that α_t is state-dependent only through the auction: it is largest exactly where P is large, so the captive tranche transfers most in the bad equilibrium. Equation (5) is the model; the rest of this section reads it.

Shape of the funding curve. Where $P \approx 0$, $\xi(d) \approx (d + wk)/R^f$ (safe pricing); where P rises, the term hP bites and ξ can *fall* — offering more face value raises less money: the “dip”, Calvo's (1988) insight; where $P \approx 1$, $\xi(d) \approx [(1 - h)d + (h + w)k]/R^f$ rises again (certain-default recovery pricing, cushioned by the captive rollover). The dip — and with it everything below — exists when

$$h[P(d) + (d - k)P'(d)] > 1 \quad \text{for some } d, \quad (7)$$

i.e., when the haircut is material and P is sufficiently steep *on the market-held tranche*: only the tranche that is repriced by beliefs can generate the dip. Under (7), ξ has a local maximum n_U followed by a local minimum n_L .

The next section presents a microfoundation for P which allows for a dip, mainly because it has a sufficiently steep region. This microfoundation assumes that $h = 1$.

Proposition 1 (existence and multiplicity). *Under Assumption 1, a temporary equilibrium exists for every admissible n_t . If (7) holds, then generically: for $n_t < n_L$ a unique equilibrium, on the first branch (low d , $p \approx 0$, spread ≈ 0 : **safe**); for $n_L < n_t < n_U$ three equilibria — G (good), U (interior, unstable), B (bad: high p , spread up to $R^f h/(1 - h)$): **fragile**; for $n_t > n_U$ only the bad one: **distressed**.*

Proposition 2 (the captive tranche and the fragile region). *Let $d_U < d_L$ denote the interior critical points of ξ (so $n_U = \xi(d_U)$, $n_L = \xi(d_L)$). By the envelope theorem,*

$$\frac{\partial n_U}{\partial k} = \frac{w + hP(d_U)}{R^f} < \frac{\partial n_L}{\partial k} = \frac{w + hP(d_L)}{R^f},$$

since $P(d_U) < P(d_L)$. Hence both edges rise with the captive stock, the lower edge faster: the fragile band $[n_L, n_U]$ shifts up and narrows as k grows, and multiplicity disappears altogether once (7) fails — beliefs can only reprice the market-held tranche, and a sufficiently captive debt stock cannot suffer a run.

Stability. Equivalently, an equilibrium is a fixed point in the conjectured object:

$$p = T(p) \equiv P\left(k + \frac{R^f(n_t - c)}{1 - hp}\right). \quad (8)$$

T is increasing; under the tâtonnement $p^{k+1} = T(p^k)$ (project at a fixed premium, add the implied spread, repeat — Blanchard’s practical iteration), G and B are stable ($T' < 1$), U is not ($T' > 1$).

Multiple equilibria. The government passively issues whatever face value covers the funding need n_t , and atomistic lenders price the issue at a conjectured clearing point that no individual lender can move. Under that protocol *every* solution of $\xi(d) = n_t$ is a rational-expectations equilibrium. Which stable equilibrium prevails is selected by beliefs, not fundamentals: the fragile region is a sunspot region. Translating its edges into inherited debt via (3), multiple equilibria are possible iff

$$d_{t-1} \in [(n_L + s_t)G_t, (n_U + s_t)G_t]. \quad (9)$$

We next develop a microfoundation for P . It comes from a model without multiple equilibria — that of CHR, in which the government *chooses* the face value issued and thereby the market’s expectation; this corresponds to the good equilibrium G . We use it to obtain n_L, n_U as functions of interpretable parameters rather than free objects.

4 The default-probability function, from first principles (CHR)

Periods t and $t+1$ are generic in this section.

4.1 The default event at $t+1$

The borrowing at period t fixes the refinancing need at $t+1$: by (3) shifted one period, $n_{t+1} = d_t/G_{t+1} - s_{t+1}$. We assume the government exerts maximum effort to avoid a default: $s_{t+1} = \bar{s}$.² Default occurs at $t+1$ iff growth comes out too low for the need to be coverable given **maximum sustainable borrowing (MSB) going forward**. The MSB, denoted b_M , is the largest *total* amount — market lending plus the captive rollover — that can be raised at $t+1$ against surpluses, extraction flows, and rollovers from $t+2$ on. Given b_M , default occurs iff

$$\frac{d_t}{G_{t+1}} - \bar{s} > b_M \quad \Longleftrightarrow \quad G_{t+1} < \frac{d_t}{\bar{s} + b_M}, \quad (10)$$

and, given b_M ,

$$P(d) = F\left(\frac{d}{\bar{s} + b_M}\right) = \Phi\left(\frac{\ln d - \ln(\bar{s} + b_M) - \mu}{\sigma}\right), \quad (11)$$

which satisfies Assumption 1.

²In the literature, this assumption is referred to as “excusable” default.

4.2 Maximum sustainable borrowing: a stationary problem

Consider a government at the start of *any* period; primes denote the following period. CHR define sustainability directly: *present borrowing is sustainable when its repayment does not rely on a sequence of future borrowing that is unbounded*. Throughout, the market tranche is priced conservatively at zero recovery (discussed at the end of the subsection); the captive tranche rolls at its price regardless.

(i) **Proceeds, given anticipated capacity.** Let b denote *total* proceeds raised now (per unit of current GDP), δ the total face value promised, and b' the total capacity anticipated for the next period. Repayment fails iff $G' < \chi \equiv \delta/(\bar{s} + b')$. Total proceeds are the market tranche at zero-recovery pricing plus the captive tranche at its price:

$$b = \frac{(1 - F(\chi))(\delta - k)}{R^f} + c = \frac{(1 - F(\chi))\chi(\bar{s} + b') + k(w + F(\chi))}{R^f}, \quad (12)$$

substituting $\delta = \chi(\bar{s} + b')$. The captive term is a net flow: the stock k is inside the obligations δ and its rollover inside the proceeds, so (12) nets the service of the captive stock automatically; what remains is the wedge w plus the default-state transfer $F(\chi)$ — the probability-weighted value the captive holders hand over by paying par on a claim that defaults with probability $F(\chi)$.

(ii) **The borrowing factor.** Maximizing the market term over the promised threshold χ defines

$$\gamma \equiv \max_{\chi} (1 - F(\chi))\chi, \quad \text{attained at } \chi_M, \quad (13)$$

a default-risk-discounted certainty-equivalent growth factor; $\gamma \leq \bar{g}$ always. Under lognormality, with $\chi = e^{\mu + \sigma z}$, the first-order condition is

$$\sigma(1 - \Phi(z_M)) = \phi(z_M), \quad \gamma = (1 - \Phi(z_M))e^{\mu + \sigma z_M}, \quad (14)$$

with a unique $z_M < 0$; at the calibration below $z_M = -2.28$.³

(iii) **The recursion, no-Ponzi, and the fixed point.** By (12), maximal total proceeds now are increasing and affine in anticipated capacity next period:

$$b = \mathcal{T}(b') \equiv \max_{\chi} \frac{(1 - F(\chi))\chi(\bar{s} + b') + k(w + F(\chi))}{R^f}. \quad (15)$$

With $\gamma < R^f$ (maintained; it holds at every calibration in this project, *including* when $R^f < \bar{g}$: the default discount $\gamma < \bar{g}$ is what restores contraction when the raw Gordon condition fails), $\mathcal{T}$ is a contraction; borrowing above its fixed point is feasible only if the sequence of future capacities explodes — a Ponzi scheme. The fixed point, to first order in k ,⁴ is

$$\boxed{b_M = \frac{\gamma \bar{s} + \alpha}{R^f - \gamma}, \quad \alpha \equiv k(w + \Phi(z_M)), \quad d_M = (\bar{s} + b_M)e^{\mu + \sigma z_M}. \quad (16)}$$

³At this z_M , Claude has also numerically verified (14).

⁴Evaluating the maximand of (15) at the baseline threshold χ_M ; the exact optimum promises slightly more (the first-order condition becomes $(1 - F) = f\chi(1 - k/\delta)$). At the calibration of Section 8 the difference is 0.4pp of GDP; the figures use the exact fixed point.

b_M is the value of a growing perpetuity of the extraction flows with growth factor γ in place of $\bar{g}$: the value of being able to roll over forever; d_M is the corresponding maximum face value. Note the exact parallel between the two horizons: the short-run realized extraction from captive creditors is $\alpha_t = k(w + h p_t)/R^f$ per period (6), its long-run counterpart is $\alpha = k(w + \Phi(z_M))$ — the same object with $h = 1$ and the default probability evaluated at the capacity margin.

Why zero recovery. Pricing future market rollovers at zero recovery is conservative twice over: it lowers b_M (CHR’s maximum-recovery extension confirms the direction, and that the magnitude is small), and it implies that, in the long run, the bad equilibrium is a sudden stop: crisis countries retain market access only in the short run.

4.3 Combining the short-run and long-run analysis

With b_M from (16), the function (11) is fully determined by country characteristics $(\bar{s}, c, w, \mu, \sigma, R^f)$. It is a smooth S-curve, essentially 0 far below d_M and rising steeply through it — the *cliff*, with steepness governed by σ . P is short-run in horizon but long-run in content: a one-period-ahead probability whose threshold embeds the entire infinite future through b_M , which is what distinguishes it from a naive “can’t pay out of next year’s surplus” measure, and why it is steep. P embeds the maintained conjecture that next period’s market functions and lends up to capacity ($b' = b_M$: the good-equilibrium continuation).

Remark 1 (pessimistic continuation). Replacing b_M by some $b < b_M$ in (11) — the extreme being a conjectured shutout — shifts P up, lowers the funding curve, and enlarges the fragile region: pessimism about future market access tightens today’s market. This is the Cole–Kehoe (2000) rollover layer; the maintained good-equilibrium continuation is the conservative, smallest-fragile-region benchmark.

4.4 Neutrality of anticipated union-wide inflation, and the role of the differential

Recall the decomposition (2). The following proposition shows that the *common* component of anticipated inflation is exactly neutral; the remark discusses the condition on the captive position that this requires; the corollary shows that what remains is the differential e .

Proposition 3 (neutrality of anticipated common inflation). *Fix $\pi \in \mathbb{R}$ and consider the shifted economy $(\tilde{\mu}, \tilde{R}^f) = (\mu + \pi, R^f e^\pi)$ — i.e., π^d and π^u both raised by π — holding $(\bar{s}, \sigma, h, w, c)$ fixed; the face value $k = c R^f / (1 + w)$ then scales with R^f . Then: (i) z_M is unchanged and $\tilde{\gamma} = \gamma e^\pi$; (ii) $\tilde{b}_M = b_M$, while $\tilde{d}_M = d_M e^\pi$; (iii) $\tilde{P}(d) = P(d e^{-\pi})$ and $\tilde{\xi}(d) = \xi(d e^{-\pi})$; (iv) the band edges n_L, n_U , the equilibrium default probabilities at any refinancing need n , and the log-spreads $\ln R - \ln R^f = -\ln(1 - hp)$ are invariant, while the equilibrium face value scales by e^π ; (v) with systematic risk (Section 5), all statements hold at every (θ_t, θ_t^c) , so the critical prices $\theta_G(n), \theta_B(n)$, the distances Δ_G, Δ_B , and Exposure are invariant.*

Proof. (i) The FOC (14) does not involve μ . (ii) In (15), substitute $\chi = \hat{\chi} e^\pi$ and $k \rightarrow k e^\pi$: since $\tilde{F}(\chi) = F(\chi e^{-\pi})$, the maximand and R^f both scale by e^π , so the fixed point b_M is unchanged; d_M scales by e^π via (16). (iii) From (11), $\tilde{P}(d) = P(d e^{-\pi})$; substituting into (5) with $k \rightarrow k e^\pi$ and c fixed gives $\tilde{\xi}(d) = \xi(d e^{-\pi})$. (iv) Every level set and extremum of $\tilde{\xi}$ in n -space coincides with that of ξ under $u = d e^{-\pi}$; the associated p is unchanged. (v) Under the kernel (17), the

$\mathbb{Q}$ -drift of the shifted economy is $(\mu - \theta\sigma) + \pi$: the shifted economy under $\mathbb{Q}$ is the original $\mathbb{Q}$ -economy shifted by π , so (i)–(iv) apply measure by measure. $\square$

Remark 2 (the indexation condition). Holding c rather than k fixed is a condition, not a bookkeeping choice. Neutrality requires that the captive sector’s *real* position be unchanged by the common shift; the object that measures the real position is the value of the rollover, c , and the face value $k = cR^f/(1+w)$ then carries nominal scale exactly as d_M does in part (ii) of the proposition. The condition is an institutional fact rather than a modeling convenience: the perimeter behind captivity is written in value-ratio terms — risk-weighted capital ratios, liquidity coverage, portfolio-share mandates — which scale with the nominal size of balance sheets, i.e., fix c . A perimeter written instead as a fixed face amount would violate neutrality: anticipated inflation would erode the captive layer’s real position, and with it α — a real effect of pure nominal scaling, produced by mis-indexed institutions. The fiscal side of the note imposes the same indexation by keeping $\bar{s}$ a share of GDP; a tax system without full indexation (bracket creep) would violate it there. The corollary below uses the condition once more: under a domestic-only shift, value-ratio regulation keeps c fixed while R^f is unchanged, so k and α are invariant to the differential e .

Corollary 1 (only the differential matters). Write $\gamma = \gamma_r e^{\pi^d}$, $R^f = R_r e^{\pi^u}$ and $\alpha = \alpha_r e^{\pi^u}$, with $\gamma_r \equiv (1 - \Phi(z_M))e^{\mu_r + \sigma z_M}$, $R_r \equiv e^r$, and $\alpha_r \equiv c R_r (w + \Phi(z_M))/(1+w)$ (the captive flow carries the scale of R^f , by Remark 2). Every real magnitude of the model depends on (π^d, π^u) only through $e = \pi^d - \pi^u$; in particular, to first order in k ,

$$b_M(e) = \frac{\gamma_r e^e \bar{s} + \alpha_r}{R_r - \gamma_r e^e}, \quad \frac{\partial b_M}{\partial e} = \frac{\gamma (\bar{s} R^f + \alpha)}{(R^f - \gamma)^2} > 0,$$

evaluated at the calibration of Section 8: $\partial b_M / \partial e \approx 6.6$ — about $6\frac{1}{2}\text{pp}$ of GDP of capacity per 100bp of expected inflation differential. The convexity comes from the small gap $R^f - \gamma$.

Proof. By Proposition 3, shift both rates by $-\pi^u$: the economy with (π^d, π^u) is real-equivalent to the economy with $(e, 0)$. Differentiate (16) using $\partial\gamma/\partial e = \gamma$ and $\partial\alpha/\partial e = 0$ (Remark 2: c is the regulated object and R^f carries π^u). $\square$

Two comments. First, the sign: for a member state, expected domestic inflation *above* the union’s raises capacity — it erodes debt denominated in the union’s unit faster than the union’s discount rate compensates. Symmetrically, expected lowflation or deflation relative to the union ($e < 0$) destroys capacity at the same rate: this is the inflation configuration that matters for euro-area credit risk. Second, a caveat: the i.i.d. assumption treats e as permanent, while in a currency union inflation differentials mean-revert through competitiveness; a calibration that reads a current differential as permanent overstates its capacity effect.

5 Systematic risk: short-run and long-run risk appetite

So far payoffs were discounted at the safe rate under physical probabilities — legitimate only if default risk is idiosyncratic. The LPPS evidence says it is not: sovereign spread changes load on US equity returns, VIX, and global risk-premium proxies; roughly one-third of the spread level is a distress risk premium; even the default-intensity component comoves with global conditions.

Default happens in global bad states, when the marginal investor's stochastic discount factor is high: it is a *systematic* risk.

Let the (log) one-factor pricing kernel load on standardized log growth $z = (\ln G - \mu)/\sigma$ with market price of risk $\theta \geq 0$:

$$M = \frac{1}{R^f} \exp(-\theta z - \frac{1}{2}\theta^2) \implies \ln G \stackrel{\mathbb{Q}}{\sim} N(\mu - \theta\sigma, \sigma^2) : \quad (17)$$

the change of measure shifts the growth drift down by $\theta\sigma$, where $\theta > 0$ iff the country's growth is low exactly when the global SDF is high.⁵ Every formula of Section 4 goes through with μ replaced by $\mu - \theta\sigma$.

We now add a specification for the market's risk appetite:

Assumption 2 (risk appetite: long-run anchor and persistence). *The current market price of distress risk θ_t is realized at the period- t auction; it reverts to a long-run anchor θ_∞ , and investors conjecture the future price relevant for continuation capacity as $\theta_t^c = \theta_\infty + \rho_\theta(\theta_t - \theta_\infty)$, $\rho_\theta \in [0, 1]$ ($\rho_\theta = 0$: risk-appetite shocks seen as transitory; $\rho_\theta = 1$: as permanent).*

The long-run risk appetite determines the maximum sustainable borrowing $b_M(\theta_t^c)$, computed from (15) under drift $\mu - \theta_t^c\sigma$ (the captive parameters (w, c) and the threshold z_M are unchanged), where $\theta_\infty > 0$ measures a degree of conservatism. The short-run risk appetite appears in the short-run default probability:

$$P_t(d) = \Phi\left(\frac{\ln d - \ln(\bar{s} + b_M(\theta_t^c)) - \mu + \theta_t\sigma}{\sigma}\right). \quad (18)$$

Using this specification, the band edges become functions of the market's risk appetite: $n_L(\theta_t)$, $n_U(\theta_t)$. These functions describe a country's "exits" from the multiple-equilibrium situation in the directions of the good and the bad equilibrium; the following section describes these exits in terms of shifts in market risk appetite.

6 Fragility in risk-appetite space: distances to the two exits

Both band edges are decreasing in θ_t : "risk-off" shifts the whole funding curve down. For a country with refinancing need n , define the two **critical risk prices**

$$\theta_G(n) : n_U(\theta_G) = n, \quad \theta_B(n) : n_L(\theta_B) = n. \quad (19)$$

The good equilibrium exists iff $\theta_t \leq \theta_G(n)$; the bad one iff $\theta_t \geq \theta_B(n)$. The country's situation is summarized by two *distances in risk-appetite units*:

$$\begin{aligned} \Delta_G = \theta_G(n_t) - \theta_t &\geq 0 : \text{ the risk-off move that would destroy the good equilibrium;} \\ \Delta_B = \theta_t - \theta_B(n_t) & : \text{ the risk-on move that would destroy the bad one.} \end{aligned} \quad (20)$$

A country is "fragile" iff $\theta_t \in [\theta_B(n_t), \theta_G(n_t)]$. Countries at the same debt position can have very different sensitivities of their funding curves to the global factor; the distances (20) are

⁵Equivalently one can keep physical probabilities and discount at $R^f + \lambda$.

denominated in units of an estimable quantity — the global risk-appetite factor — hence comparable across countries and convertible into probabilities:

$$\text{Exposure}_{it} = \mathbb{P}[\theta > \theta_G(n_{it})] \quad (21)$$

is the probability of a risk-appetite realization that leaves the country with only the bad equilibrium — a well-defined tail probability, not a heuristic index.

Persistence is key. By (18), θ_t enters twice: in the one-period drift (a small effect) and in the continuation capacity $b_M(\theta_t^c)$ (the large effect). With $\rho_\theta = 0$ — risk-off seen as transitory — capacity is untouched and the fragile region barely moves (at the calibration below, $\theta_t = 0.5$ moves n_U from 51.9% only to 51.1%); with $\rho_\theta = 1$ it moves to 43.4%. *A risk-off episode is dangerous in proportion to how permanent the market believes it to be* — the asset-pricing analogue of Blanchard’s point that permanent shifts in $r - g$, not transitory ones, are what threaten debt sustainability; consistent with LPPS’s finding that risk-neutral intensity dynamics are more persistent than physical ones.

Hysteresis. Suppose markets coordinate on G whenever it exists; outcomes are then deterministic in (n_t, θ_t) with a discontinuous repricing at $\theta_t = \theta_G(n_t)$ — and hysteresis: once in B , returning to G requires not $\theta_t < \theta_G$ but the much stronger $\theta_t < \theta_B(n_t)$, or a backstop.

Backstop. Inside the fragile region any selection process $\zeta_t \in \{G, B\}$ is admissible — the indeterminacy survives at known fundamentals, which is why a credible backstop *announcement* (“whatever it takes”) can delete B .

7 Two limits that recover the parent models

Limit 1 ($\sigma \rightarrow 0$, $k = 0$): **Blanchard’s box.** $P(d) \rightarrow \mathbf{1}\{d > d_M\}$: the funding curve becomes two straight lines with a vertical drop at d_M ; the fragile region converges to a band of Blanchard’s (2022, ch. 4.5) form — lower edge proportional to $(1 - h)$ — with one upgrade: his free threshold b^* is now d_M , a function of $(\bar{s}, \mu, \sigma, R^f, \theta_\infty)$: the band’s edges are made of observables.

Limit 2 ($h \rightarrow 1$, $\theta_t \rightarrow \theta_\infty$, $k = 0$): **CHR.** With zero recovery, the bad equilibrium is a loss of market access: CHR’s excusable-default sudden stop, and their debt Laffer curve is exactly ξ with $h = 1$, $k = 0$. Partial recovery $h < 1$ (in the short run) is what separates “bad equilibrium” from “no market access,” creating the expensive-but-priced branch on which Greece 2011 and Portugal 2011–12 briefly sat.

8 Calibrated illustration

Annual calibration for a typical euro-area member: $\mu = 3\%$, $\sigma = 3\%$, $\bar{s} = 4\%$, $R^f = 1.03$, $h = 30\%$, plan $s = 2\%$; captive stock $k = 15\%$ of GDP with $w = 0$ (regulatory home bias of domestic banks and mandated funds; order of magnitude of euro-area domestic-bank holdings of own-sovereign debt), benchmark risk pricing $\theta_t = \theta_\infty = 0$ (Figure 1), persistence $\rho_\theta = 1$

One calibration, three views ($\mu=3\%$, $\sigma=3\%$, $\bar{s}=4\%$, $R^f=1.03$, $h=30\%$, $k=15\%$, $w=0$; $b_M=51\%$, $d_M=53\%$ of GDP; $\theta_t=\theta_\infty=0$)

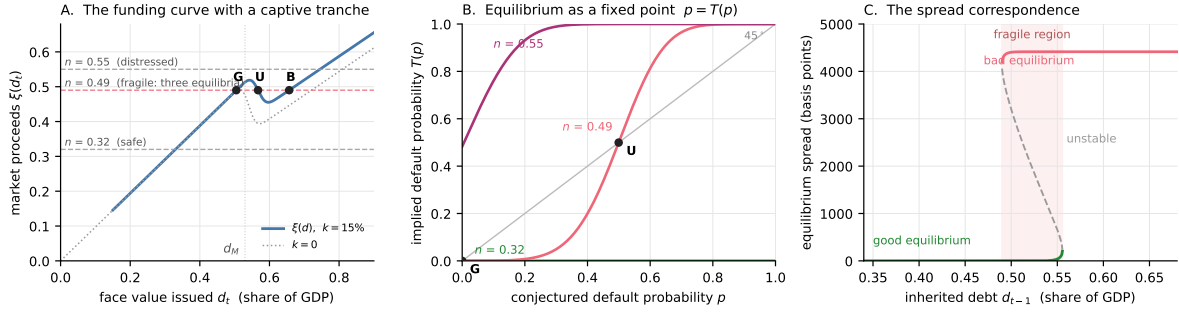


Figure 1: Three views of the same equilibrium at the risk-neutral benchmark ($\theta_t = \theta_\infty = 0$), with the captive tranche ($k = 15\%$, $w = 0$). *Panel A*: the funding curve $\xi(d)$ with three refinancing-need lines; the dotted curve is the $k = 0$ benchmark — the dip narrows because only the market-held tranche is repriced. *Panel B*: the same equilibria as fixed points of the belief map $T(p)$. *Panel C*: the equilibrium spread correspondence over inherited debt.

where θ varies (Figure 2). Then $\gamma = 0.951$ and $\alpha = \Phi(z_M)k = 0.17\%$ of GDP — a small flow whose capitalization is not small: $b_M = 51.0\%$ of GDP against 48.5% at $k = 0$ (first-order formula: 50.6%), with $d_M = 53.0\%$. The band edges are $n_U = 51.9\%$, $n_L = 45.5\%$: relative to $k = 0$ (49.2% and 39.4%), both edges rise and the fragile band *narrows* from 9.8 to 6.4pp of GDP (Proposition 2; the envelope derivatives at the calibration are $\partial n_U / \partial k = 0.02$ and $\partial n_L / \partial k = 0.28$). In inherited-debt space the fragile band is $[49\%, 56\%]$ of GDP, against $[43\%, 53\%]$ at $k = 0$. Good-equilibrium spreads: a few bp to ~ 100 bp; bad-equilibrium: $\sim 1,900$ – $4,400$ bp. Conservative long-run pricing lowers capacity: $\theta_\infty = 0.3$ gives $b_M = 45.7\%$, $\theta_\infty = 0.5$ gives 42.7% . In Figure 2, a country with $n_t = 40\%$ has $\theta_B = 0.39$ and $\theta_G = 0.76$: fragile for $\theta_t \in [0.39, 0.76]$, with exit distances read off the horizontal axis. A cliff-side feature worth advertising for the empirics: because P is steep, a moderate θ produces enormous ratios of risk-neutral to physical default probabilities just below the limit ($d = 47\%$: $P \approx 0$ under $\theta = 0$ but ≈ 0.38 under $\theta = 0.5$) — the “visible spreads, near-zero historical default frequency” configuration advanced sovereigns actually present.

Remark 3 (what the spread correspondence means empirically). Over the fragile region the market spread is not a function of fundamentals: two spreads are consistent with the same (n_t, θ_t) . A fragile country can be repriced violently with no fiscal news — and, by Section 6, the repricing is triggered by a *measurable* variable, the global risk-appetite factor, in proportion to its perceived persistence.

9 Deliberate omissions

Three classic omissions, each a documented choice with a natural owner in the literature: no strategic default (ability-to-pay only); no maturity choice (one-period debt — the maturity structure of actual issuance slows the bad equilibrium into a slow-moving crisis, Lorenzoni–Werning 2019, and re-enters through the financing-need identity of Appendix B); no investor heterogeneity beyond the one-factor kernel.

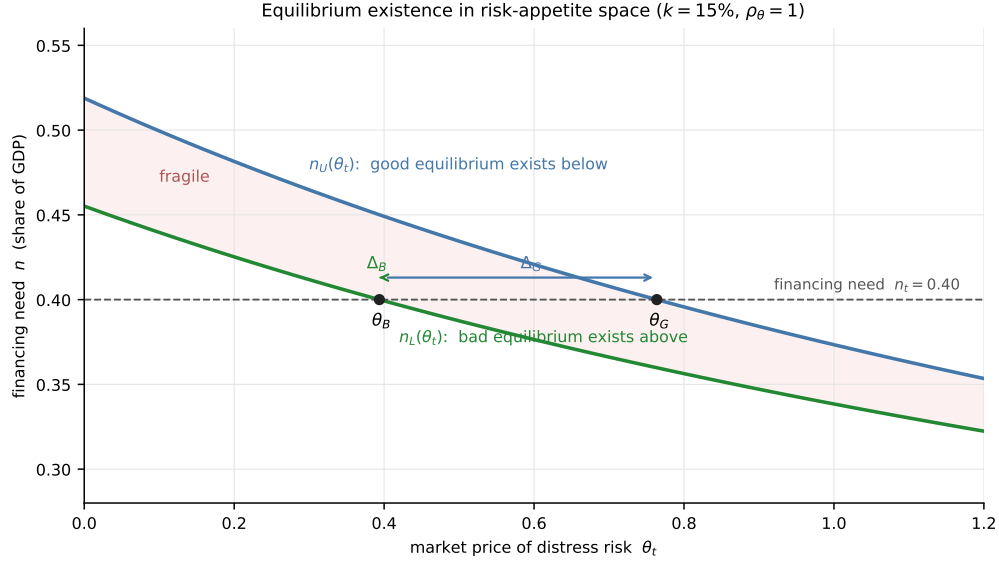


Figure 2: Equilibrium existence in risk-appetite space ($k = 15\%$, $\theta_\infty = 0$, $\rho_\theta = 1$). The good equilibrium exists below $n_U(\theta_t)$, the bad one above $n_L(\theta_t)$. A country with need $n_t = 0.40$ is fragile for $\theta_t \in [\theta_B, \theta_G] = [0.39, 0.76]$; its exit distances Δ_B and Δ_G are measured along the horizontal axis and, given the estimated factor distribution, convert to the exposure probability (21).

Three further blocks were developed in earlier drafts and are removed from the essentials, for stated reasons. First, a *robustness* reading of θ : the kernel (17) is formally identical to a Hansen–Sargent worst-case drift distortion, so θ can be read as the size of a set of models the marginal lender entertains, with θ_∞ bounded by a detection-error probability rather than by preference arguments. The reading is observationally equivalent in prices; it is removed because the entertained model set is hard to specify and calibrate. Second, *state-dependent institutions*: the captive layer is itself a creditor of the sovereign, so in stress its flow can stop or reverse (the sovereign–bank loop, Brunnermeier et al. 2016; Farhi–Tirole 2018). Here only the auction channel is kept — α_t varies with p_t , and is largest in the bad equilibrium — while the continuation parameters (w, c) are constants; the loop’s within-period fixed point is excluded, which is what keeps the funding curve (5) one-dimensional. Third, a possible third extraction component — a dilutable *payment-unit* layer (seigniorage-like, the junior claim in Bolton–Huang’s sense) — is left undecided: it is less well worked out than the captive-creditor component, and the demarcation between creditors and payment-system operators is not trivial.

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A Notation map

The note’s notation is independent; the table maps it to the three source papers and to the Commission-DSA replication code. A dash means the concept is absent from that source. In the stationary capacity problem of Section 4.2, primes denote the following period and δ the face value promised — calendar indices are reserved for the short-run analysis. Two cautions on symbols: the recursion operator of (15) is $\mathcal{T}$; and the note’s α (the captive extraction flow) is unrelated to CHR’s α (their symbol for the maximum surplus) and to the repository’s maturity share α^{ST} . The captive-creditor symbols $(c, w, \bar{c}, k, \alpha_t, \alpha)$ have no counterpart in the sources, except that α corresponds to what Reinhart–Sbrancia measure as annual liquidation revenue.

Concept	Note	CHR (2015)	Blanchard (2022)	LPPS (2011)	DSA repo
gross GDP growth	G_{t+1}, G'	g_{t+1} (gross)	$1 + g$ ($g=0$ in box)	—	$1 + g_t$
log-growth mean, vol	μ, σ	μ, σ	—	—	quarterly moments
inflation decomposition	π^d, π^u, e	—	—	—	(inflation target)
gross safe return	R^f	$1 + r$	$1 + r$	riskless curve r	— (see market rates)
gross promised return	R_t	implicit $R(d)$	$1 + R$	—	$1 + i^{ST}, 1 + i^{LT}$ (exog.)
spread	ψ_t	yield spread ω	$(1+r)px/(1-px)$	CDS spread	—
max primary surplus	$\bar{s}$	α	$s_{\max}$	—	— (no ceiling)
planned primary surplus	s_t	($= \alpha$)	s	—	pb_t, spb_t
captive proceeds, wedge	$c, w; \bar{c}(w)$	—	—	—	—
captive face value (short-hand)	k	$=$ —	—	—	—
extraction flow (captive)	$cR^f/(1+w)$	—	—	—	—
face value due / GDP	$\alpha_t; \alpha$	—	—	—	(in iir_t)
face value due / GDP	$d_t; \delta$ (stationary)	d_t	b	—	d_t (debt ratio)
borrowing proceeds / GDP	b, b' (total)	b_t	—	—	—
refinancing need	n_t	—	—	—	GFN_t
default growth threshold	χ	x	—	—	—
borrowing factor	γ	γ	—	—	—
max sustainable borrowing	b_M	b_M	—	—	—
max sustainable debt / limit	d_M	d_M	b^* (free param.)	—	—
one-period default prob.	$p_t; P(d), P_t(d)$	$PD(d), F(x)$	$p \in \{0, 1\}$	intensity λ_t	—
haircut (LGD)	h	— (zero recovery)	x	$1 - R^Q, R^Q=0.25$	—
price of distress risk	θ_t	—	—	$\eta_t; \mathbb{P}$ -vs- $\mathbb{Q}$ dyn.	—
long-run anchor, persistence	$\theta_\infty, \rho_\theta$	—	—	($\mathbb{Q}$ more persistent)	—
pricing kernel	M	—	—	implicit in $\mathbb{Q}$	—
funding curve	$\xi(d)$	proceeds $B(d)$	—	—	—
band edges; critical prices	$n_L, n_U; \theta_B, \theta_G$	—	multiplicity range	—	—
selection (sunspot)	ζ_t	excluded (fn. 3)	sunspot	—	—
maturity shares	(App. B)	—	—	—	α^{ST}, β

Table 1: Mapping of the note’s notation to the sources. CHR: Collard–Habib–Rochet (JEEA 2015). Blanchard: *Fiscal Policy under Low Interest Rates* (2022), ch. 4.5 box. LPPS: Longstaff–Pan–Pedersen–Singleton (AEJ:Macro 2011), Pan–Singleton pricing. DSA repo: Darvas–Welslau–Zettelmeyer replication of the Commission DSA.

B Relation to the Commission DSA

The model is self-standing; this appendix records how it meets operational practice. The Commission DSA (as replicated in the Bruegel/Welslau repository) projects the debt ratio via

$$d_t = d_{t-1} \frac{1 + iir_t}{1 + g_t} - pb_t + sfa_t, \quad (22)$$

with the implicit interest rate a maturity-weighted moving average of past market rates,

$$iir_t^{LT} = \beta_{t-1} i_t^{LT} + (1 - \beta_{t-1}) iir_{t-1}^{LT}, \quad iir_t = \alpha_{t-1}^{ST} i_t^{ST} + (1 - \alpha_{t-1}^{ST}) iir_t^{LT}, \quad (23)$$

where the market rates i^{ST}, i^{LT} are *exogenous*: Bloomberg forwards to $T+10$, then linear convergence to 2% plus the inflation target — an assumption of zero long-run intra-union spreads, i.e., an invisible selection of the good equilibrium forever. (By Proposition 3, the terminal point “2% plus the inflation target” imposes the neutrality of anticipated inflation by construction in the long run; the proposition states when that convention is innocuous.) The model determines exactly the object this convention fixes: the market rate on newly issued debt. Note also that (22) rearranged is the refinancing-need identity (3), and that the repository’s maturity machinery (α^{ST} , β , redemption shares) is what converts the model’s one-period repricing into a realistic slow-then-sudden pass-through.

Two measurement notes. The extraction flow from captive creditors does not appear in pb_t : it appears in the *interest line*, as the wedge between iir_t and market rates, times the captive stock — which is precisely how Reinhart–Sbrancia measure liquidation revenue, so the model’s definition of α_t and its data counterpart agree. And n_t is the published gross financing need, so every input of the model is fiscal or market data.